

2201NSC Linear Algebra and Applications

Week 3

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Determinants, Inverses and Cramer's Rule

A 1×1 determinant is its only entry

For a 1×1 matrix

$$A = (a),$$

the determinant is defined by

$$\det(A) = a.$$

Example

$$A = (-3) \quad \implies \quad \det(A) = -3.$$

The determinant is the entry itself, not its absolute value.

The 2×2 determinant is a diagonal difference

$$\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

Example

$$\begin{vmatrix} 4 & 7 \\ 2 & 6 \end{vmatrix} = (4)(6) - (7)(2) = 24 - 14 = 10.$$

Multiply down the main diagonal, then **subtract** the other diagonal product.

Expand a 3×3 determinant along the first row

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

Using the first-row sign pattern $+, -, +$,

$$\begin{aligned} \det(A) &= 1 \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} - 2 \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} + 3 \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} \\ &= 1(45 - 48) - 2(36 - 42) + 3(32 - 35) = -3 + 12 - 9 = 0. \end{aligned}$$

How each minor is formed

For each entry in the first row, delete that entry's row and column. Write the remaining four entries as the determinant of a 2×2 .

Determinant properties often remove the arithmetic

Structure or operation	Effect on the determinant
A zero row or column	$\det(A) = 0$
Two equal rows or columns	$\det(A) = 0$
One row/column is a combination of others	$\det(A) = 0$
A is triangular	Product of the diagonal entries
Swap two rows	Changes the sign
Multiply one row by k	Multiplies the determinant by k

Spot $\det(A) = 0$ before expanding

Repeated rows

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 1 & 2 & 3 \end{pmatrix}$$

$$\det(A) = 0$$

Proportional rows

$$\begin{pmatrix} 1 & -2 & 3 \\ 2 & -4 & 6 \\ 0 & 1 & 5 \end{pmatrix}$$

$$\det(A) = 0$$

One row is a sum

$$\begin{pmatrix} 1 & 0 & 2 \\ 3 & 1 & -1 \\ 4 & 1 & 1 \end{pmatrix}$$

$$R_1 + R_2 = R_3$$

$$\det(A) = 0$$

A structural reason is faster than a full expansion.

Row operations predict how the determinant changes

Row operation

$$R_i \leftrightarrow R_j$$

$$R_i \leftarrow kR_i$$

$$R_i \leftarrow R_i + kR_j$$

Effect

changes sign

multiplies by k

unchanged

Inverse Matrices

Let $A \in \mathbb{R}^{n \times n}$. An **inverse** of A is a matrix A^{-1} such that

$$AA^{-1} = A^{-1}A = I_n.$$

- ▶ Only **square matrices** can have an inverse.
- ▶ The inverse A^{-1} has the same dimensions as A .
- ▶ Not every square matrix is invertible.

The determinant determines

$$A^{-1} \text{ exists} \iff \det(A) \neq 0.$$

For square matrices, these statements are equivalent

$$\det(A) \neq 0 \iff A^{-1} \text{ exists}$$

$$\iff A\mathbf{x} = \mathbf{b} \text{ has a unique solution for every } \mathbf{b}$$

$$\iff A \text{ has a pivot in every row and column.}$$

If $\det(A) = 0$

A is **singular**: it has no inverse, and $A\mathbf{x} = \mathbf{b}$ may have no solution or infinitely many solutions.

A 2×2 inverse exists only when $ad - bc \neq 0$

For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Example

$$A = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}, \quad \det(A) = 8 - 3 = 5, \quad A^{-1} = \frac{1}{5} \begin{pmatrix} 4 & -1 \\ -3 & 2 \end{pmatrix}.$$

The formula cannot be used when $ad - bc = 0$ because the inverse does not exist.

Gauss–Jordan elimination to find A^{-1}

$$(A \mid I) \longrightarrow (I \mid A^{-1}).$$

1. Form the augmented matrix $(A \mid I)$.
2. Apply valid row operations.
3. If the left side becomes I , the right side is A^{-1} .
4. If a pivot is missing, A is singular and no inverse exists.

Quick check

Multiply the result by the original matrix and confirm that $AA^{-1} = I$.

Cramer's rule

For $A\mathbf{x} = \mathbf{b}$ with $\det(A) \neq 0$,

$$x_i = \frac{\det(A_i)}{\det(A)},$$

where A_i is the matrix A with column i replaced by $\mathbf{b}$.

Three things to remember

- ▶ Replace a **column**, not a row.
- ▶ The denominator is always $\det(A)$.
- ▶ If $\det(A) = 0$, Cramer's rule cannot be used.

Worked example: Cramer's rule for two equations

$$2x + y = 5,$$

$$x - y = 1.$$

$$A = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, \quad \det(A) = -3.$$

$$x = \frac{\begin{vmatrix} 5 & 1 \\ 1 & -1 \end{vmatrix}}{-3} = \frac{-6}{-3} = 2,$$

$$y = \frac{\begin{vmatrix} 2 & 5 \\ 1 & 1 \end{vmatrix}}{-3} = \frac{-3}{-3} = 1.$$

Check

$2(2) + 1 = 5$ and $2 - 1 = 1$, so $(x, y) = (2, 1)$ satisfies both equations.

Short Revision Quiz

Determinants, inverses, Cramer's rule and fast structural reasoning

Question 1 — True or false

If $\det(A) \neq 0$, then $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b}$.

A. True

B. False

Question 2 — True or false

If $A\mathbf{x} = \mathbf{b}$ is inconsistent, then A^{-1} exists.

A. True

B. False

Question 3 — True or false

If $\det(A) = 0$, then $A\mathbf{x} = \mathbf{b}$ must have infinitely many solutions.

A. True

B. False

Question 4 — Zero determinant at a glance

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 1 & 2 & 3 \end{pmatrix}.$$

Why is $\det(A) = 0$? Explain without expanding the determinant.

Question 5 — Zero determinant at a glance

$$B = \begin{pmatrix} 1 & -2 & 3 \\ 2 & -4 & 6 \\ 0 & 1 & 5 \end{pmatrix}.$$

Why is $\det(B) = 0$? Explain without expanding the determinant.

Question 6 — Zero determinant at a glance

$$C = \begin{pmatrix} 2 & 1 & -3 \\ 0 & 0 & 0 \\ 4 & 5 & 6 \end{pmatrix}.$$

Why is $\det(C) = 0$? Explain without expanding the determinant.

Question 7 — Zero determinant at a glance

$$D = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 0 & 3 \\ 0 & 0 & 4 \end{pmatrix}.$$

Why is $\det(D) = 0$? Explain without expanding the determinant.

Question 8 — Invertibility

$$A = \begin{pmatrix} k & 2 \\ 3 & 1 \end{pmatrix}.$$

For which value of k does A fail to have an inverse?

- A. $k = 0$
- B. $k = 2$
- C. $k = 4$
- D. $k = 6$

Question 9 — True or false

The inverse A^{-1} exists if and only if $\det(A) \neq 0$.

A. True

B. False

Question 10 — Calculate a determinant

Calculate the determinant of

$$A = \begin{pmatrix} 3 & 2 \\ 1 & 4 \end{pmatrix}.$$

Quiz answers 1–5

Q	Answer	Reason
1	True	A is invertible, so $\mathbf{x} = A^{-1}\mathbf{b}$ is unique for every $\mathbf{b}$.
2	False	If A^{-1} existed, every $\mathbf{b}$ would give a unique solution.
3	False	A singular system may have no solution or infinitely many solutions.
4	Equal rows	Rows 1 and 3 are identical, so the rows are dependent.
5	Proportional rows	Row 2 is twice row 1, so the rows are dependent.

Quiz answers 6–10

Q	Answer	Reason
6	Zero row	A matrix with a zero row has determinant zero.
7	Zero diagonal entry	Since D is triangular, $\det(D) = 2 \cdot 0 \cdot 4 = 0$.
8		$\det(A) = k - 6$, so A is singular when $k = 6$.
9	True	Nonzero determinant and invertibility are equivalent.
10	10	$\det(A) = (3)(4) - (2)(1) = 12 - 2 = 10$.